Module 11: Switch Security Configuration

Switching, Routing and Wireless Essentials v7.0 (SRWE)

Module Objectives

Module Title: Switch Security Configuration

Module Objective: Configure switch security to mitigate LAN attacks

Topic Title	Topic Objective
Implement Port Security	Implement port security to mitigate MAC address table attacks.
Mitigate VLAN Attacks	Explain how to configure DTP and native VLAN to mitigate VLAN attacks.
Mitigate DHCP Attacks	Explain how to configure DHCP snooping to mitigate DHCP attacks.
Mitigate ARP Attacks	Explain how to configure ARP inspection to mitigate ARP attacks.
Mitigate STP Attacks	Explain how to configure PortFast and BPDU Guard to mitigate STP Attacks.

11.1 Implement Port Security

Secure Unused Ports

Layer 2 attacks are some of the easiest for hackers to deploy but these threats can also be mitigated with some common Layer 2 solutions.

- All switch ports (interfaces) should be secured before the switch is deployed for production use.
- How a port is secured depends on its function.
- Disable all unused ports on a switch.
- Use the IOS **shutdown** command.
- If a port must be reactivated at a later time, it can be enabled with the **no shutdown** command.
- To configure a range of ports, use the **interface range** command.

```
Switch(config)# interface range type module/first-number – last-number
```

Secure Unused Ports

- Show the status of the interfaces
- This displays the list of interfaces and whether they are disabled

```
Switch# show interface status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        disabled 1          auto    auto  10/100BaseTX
Fa0/2                        notconnect   1          auto    auto   10/100BaseTX
Fa0/3                        notconnect   1          auto    auto   10/100BaseTX
Fa0/4                        notconnect   1          auto    auto   10/100BaseTX
Fa0/5                        notconnect   1          auto    auto   10/100BaseTX
Fa0/6                        notconnect   1          auto    auto   10/100BaseTX
Fa0/7                        notconnect   1          auto    auto   10/100BaseTX
Fa0/8                        notconnect   1          auto    auto   10/100BaseTX
Fa0/9                        notconnect   1          auto    auto   10/100BaseTX
Fa0/10                       notconnect   1          auto    auto   10/100BaseTX
Fa0/11                       notconnect   1          auto    auto   10/100BaseTX
```

Secure Unused Ports

- Use the interface range command if appropriate
- Shutdown all the ports

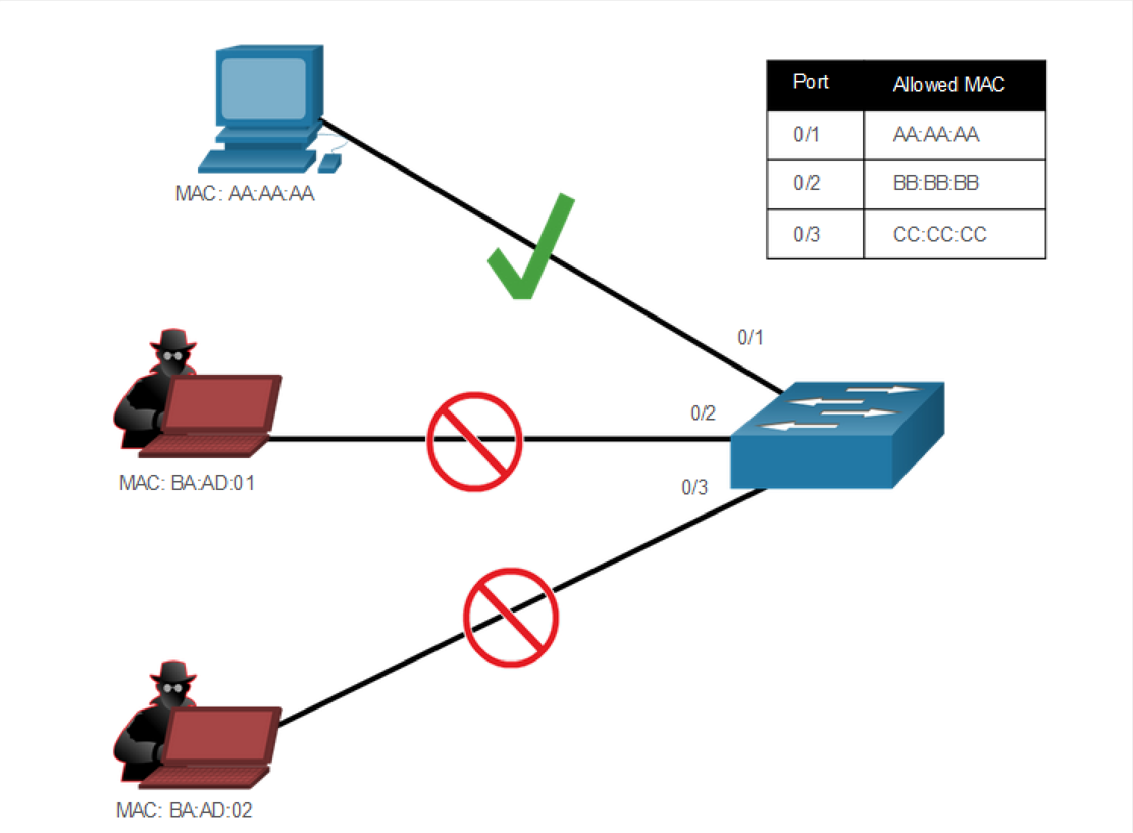
```
S1(config)# interface range fa0/8 - 24
S1(config-if-range)# shutdown
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
(output omitted)
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
S1(config-if-range)#
```

Mitigate MAC Address Table Attacks

The simplest and most effective method to prevent MAC address table overflow attacks is to enable port security.

Port security limits the number of valid MAC addresses allowed on a port

- Manually configure the addresses
- Dynamically learn a limited number of MAC addresses



Enable Port Security

Port security is enabled with the **switchport port-security** interface configuration command.

The port must me manually configured as an access port

```
S1(config)# interface f0/1
S1(config-if)# switchport port-security
Command rejected: FastEthernet0/1 is a dynamic port.
S1(config-if)# switchport mode access
S1(config-if)# switchport port-security
S1(config-if)# end
S1#
```

Enable Port Security (Cont.)

Use the **show port-security interface** command to display the current port security settings for FastEthernet 0/1.

- violation mode is shutdown
- maximum number of MAC addresses is 1
- If more than one device is connected to that port, the port will transition to the error-disabled state.

```
S1# show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
S1#
```

Enable Port Security (Cont.)

After port security is enabled, other port security specifics can be configured, as shown in the example.

```
S1(config-if)# switchport port-security ?
  aging         Port-security aging commands
  mac-address   Secure mac address
  maximum       Max secure addresses
  violation     Security violation mode
  <cr>
S1(config-if)# switchport port-security
```

Limit and Learn MAC Addresses

To set the maximum number of MAC addresses allowed on a port, use the following command:

```
Switch(config-if)# switchport port-security maximum value
```

```
S1(config)# interface f0/1
S1(config-if)# switchport port-security maximum ?
  <1-8192>  Maximum addresses
S1(config-if)# switchport port-security maximum
```

Limit and Learn MAC Addresses (Cont.)

The switch can be configured to learn about MAC addresses on a secure port in one of three ways:

1. Manually Configured: The administrator manually configures a static MAC address(es) by using the following command for each secure MAC address on the port:

```
Switch(config-if)# switchport port-security mac-address mac-address
```

2. Dynamically Learned: When the **switchport port-security** command is entered, the current source MAC for the device connected to the port is automatically secured but is not added to the running configuration. If the switch is rebooted, the port will have to re-learn the device's MAC address.

3. Dynamically Learned – Sticky: The administrator can enable the switch to dynamically learn the MAC address and "stick" them to the running configuration by using the following command:

```
Switch(config-if)# switchport port-security mac-address sticky
```

Saving the running configuration will commit the dynamically learned MAC address to NVRAM.

Implement Port Security

Limit and Learn MAC Addresses (Cont.)

The example demonstrates a complete port security configuration for FastEthernet 0/1.

- The administrator specifies a maximum of 4 MAC addresse
- Manually configures one secure MAC address
- Configures the port to dynamically learn additional secure MAC addresses up to the 4 secure MAC address maximum.
- Use the **show port-security interface** and the **show port-security address** command to verify the configuration.

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport port-security
S1(config-if)# switchport port-security maximum 4
S1(config-if)# switchport port-security mac-address aaaa.bbbb.1234
S1(config-if)# switchport port-security mac-address sticky
S1(config-if)# end
S1# show port-security interface fa0/1
Port Security               : Enabled
Port Status                 : Secure-up
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 4
Total MAC Addresses         : 1
Configured MAC Addresses    : 1
Sticky MAC Addresses        : 0
Last Source Address:Vlan    : 0000.0000.0000:0
Security Violation Count    : 0
S1# show port-security address
               Secure Mac Address Table
-----------------------------------------------------------------------------
Vlan    Mac Address       Type                          Ports   Remaining Age
                                                                   (mins)
----    -----------       ----                          -----   -------------
   1    aaaa.bbbb.1234    SecureConfigured              Fa0/1         -
-----------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 8192

S1#
```

Port Security Violation Modes

If the MAC address of a device attached to a port differs from the list of secure addresses, then a port violation occurs and the port enters the error-disabled state.

- To set the port security violation mode, use the following command:

```
Switch(config-if)# switchport port-security violation {shutdown | restrict | protect}
```

The following table shows how a switch reacts based on the configured violation mode.

Mode	Description
shutdown (default)	The port transitions to the error-disabled state immediately, turns off the port LED, and sends a syslog message. It increments the violation counter. When a secure port is in the error-disabled state, an administrator must re-enable it by entering the **shutdown** and **no shutdown** commands.
restrict	The port drops packets with unknown source addresses until you remove a sufficient number of secure MAC addresses to drop below the maximum value or increase the maximum value. This mode causes the Security Violation counter to increment and generates a syslog message.
protect	This is the least secure of the security violation modes. The port drops packets with unknown MAC source addresses until you remove a sufficient number of secure MAC addresses to drop below the maximum value or increase the maximum value. No syslog message is sent.

Port Security Violation Modes (Cont.)

The example shows an administrator changing the security violation to “Restrict”.

The output of the **show port-security interface** command confirms that the change has been made.

```
S1(config)# interface f0/1
S1(config-if)# switchport port-security violation restrict
S1(config-if)# end
S1#
S1# show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 4
Total MAC Addresses        : 1
Configured MAC Addresses   : 1
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0050.56be.e4dd:1
Security Violation Count   : 1
S1#
```

Ports in error-disabled State

When a port is shutdown and placed in the error-disabled state, no traffic is sent or received on that port.

A series of port security related messages display on the console, as shown in the following example.

Note: The port protocol and link status are changed to down and the port LED is turned off.

```
*Sep 20 06:44:54.966: %PM-4-ERR_DISABLE: psecure-violation error detected on Fa0/18, putting Fa0/18 in
err-disable state
*Sep 20 06:44:54.966: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC
address 000c.292b.4c75 on port FastEthernet0/18.
*Sep 20 06:44:55.973: %LINEPROTO-5-PPDOWN: Line protocol on Interface FastEthernet0/18, changed state
to down
*Sep 20 06:44:56.971: %LINK-3-UPDOWN: Interface FastEthernet0/18, changed state to down
```

Ports in error-disabled State (Cont.)

show interface

The command identifies the port status as **err-disabled**

show port-security interface command now shows the port status as **secure-shutdown**

To re-enable the port, first use **shutdown** command then use the **no shutdown** command.

```
S1# show interface fa0/18
FastEthernet0/18 is down, line protocol is down (err-disabled)
(output omitted)
S1# show port-security interface fa0/18
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 1
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : c025.5cd7.ef01:1
Security Violation Count   : 1
S1#
```

Verify Port Security

Verify that the port security is set correctly, and check to ensure that the static MAC addresses have been configured correctly.

To display port security settings for the switch, use the **show port-security** command.

```
S1# show port-security
Secure Port  MaxSecureAddr  CurrentAddr  SecurityViolation  Security Action
                (Count)        (Count)          (Count)
---------------------------------------------------------------------------
      Fa0/1              1            0                  0         Shutdown
      Fa0/2              1            0                  0         Shutdown
      Fa0/3              1            0                  0         Shutdown
(output omitted)
     Fa0/24              1            0                  0         Shutdown
---------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 4096
Switch#
```

Verify Port Security (Cont.)

Use the **show port-security interface** command to view details for a specific interface, as shown previously and in this example.

```
S1# show port-security interface fastethernet 0/18
Port Security               : Enabled
Port Status                 : Secure-up
Violation Mode              : Shutdown
Aging Time                  : 0 mins
Aging Type                  : Absolute
SecureStatic Address Aging  : Disabled
Maximum MAC Addresses       : 1
Total MAC Addresses         : 1
Configured MAC Addresses    : 0
Sticky MAC Addresses        : 0
Last Source Address:Vlan    : 0025.83e6.4b01:1
Security Violation Count    : 0
S1#
```

Verify Port Security (Cont.)

To verify that MAC addresses are "sticking" to the configuration, use the **show run** command as shown in the example for FastEthernet 0/19.

```
S1# show run | begin interface FastEthernet0/19
interface FastEthernet0/19
switchport mode access
switchport port-security maximum 10
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0025.83e6.4b02
(output omitted)
S1#
```

Verify Port Security (Cont.)

To display all secure MAC addresses that are manually configured or dynamically learned on all switch interfaces

show port-security address command as shown in the example.

```
S1# show port-security address
Secure Mac Address Table
-----------------------------------------------------------------------------
Vlan    Mac Address       Type             Ports      Remaining Age
                                                         (mins)
----    -----------       ----             -----      -------------
1       0025.83e6.4b01    SecureDynamic    Fa0/18           -
1       0025.83e6.4b02    SecureSticky     Fa0/19           -
-----------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 8192
S1#
```